Spring 2025	Machine Learning HW 1	Due: Feb 21

Problem 1 [20 points] Linear Regression I

We would like to fit a linear regression estimate to the dataset with X and Y two variables, and $Y = wX$ where w is a constant showing the weight. Given$n$ independent sample pairs, $(x^{(1)}, y^{(1)})$, $(x^{(2)}, y^{(2)})$,..., $(x^{(n)}, y^{(n)})$, we aim to estimate w by solving the following problem.

$$w^* = \arg\min_w \frac{1}{2}\left(\sum_{i=1}^{n}(y^{(i)} - wx^{(i)})^2 + \lambda w^2\right)$$

where $\lambda \geq 0$ is a tuning hyperparameter.

a) Give a solution in explicit formula for w^* so that w^* can be computed using only the training data and λ. Hint: take the derivative of the objective function and set it to zero.

b) When λ goes from 0 to infinity, how does w^* change? Give a brief explanation of your answer.

Problem 2 [20 points] Linear Regression II

Suppose you know the number of keyboard and mic sold at various locations around the world and from that you want to estimate the number of computers sold using linear regression. Your model is $Y = w_1 k + w_2 m$ where Y is the number of computers sold, k is the number of keyboards sold and m is the number of mice sold. You get 101 observations such that 100 of them have 1 keyboard, 1 mouse, and 1 computer, but the 101st has 1 keyboard, 0 mouse, and 1 computer.

For (a) and (b), you can use `LinearRegression()` in Scikit-learn to computer the answers (turn off the intercept in linear model: fit_intercept = False): link

a) What are the optimal values of w_1 and w_2 in the scenario above?

b) Now suppose you get two additional observations, both with 0 keyboard, 1 mouse, and 1 computer. What are the optimal w_1 and w_2 values now?

c) As you should notice, the optimal values for w fluctuate wildly with the addition of even very few observations. Why is this problem happening? Given an arbitrary dataset X, Y, how can we test whether such behavior might occur?

Problem 3 [30 points] Logistic Regression

For logistic regression model, we first have a decision function $z = b + w_1x_1 + w_2x_2 + ... + w_nx_n$; and we have an object assigned to positive example with a probability of $p(y = 1|\mathrm{x}) = \sigma(z) = \frac{1}{1+e^{-z}}$.

Consider the following 15 points in a two-dimensional feature space (x_1, x_2) with labels as indicated (green + is positive, red × is negative):

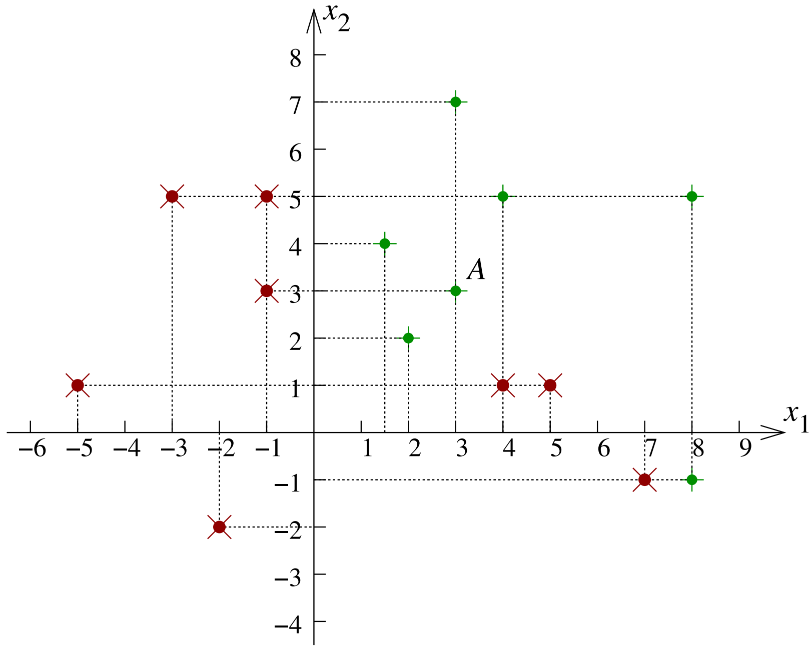


Figure 1: Data set for question 2

Given the decision function

$$z(x_1, x_2) = 4x_1 + 6x_2 - 24$$

answer the following questions:

a) What is the z value of the decision function for point A? And what's the probability of point A as a positive example?

b) In the figure above, draw the decision boundary, $z(x_1, x_2) = 0$, and indicate the positive and negative regions (half-planes).

c) What is the accuracy of this linear classifier?

d) Write the equation for a linear decision function that improves upon this accuracy.

Problem 4 [30 points] Model Evaluation

a) What are the possible reasons for model overfitting?

b) What can you do to avoid overfitting? Elaborate you answers with **logistic regression**.

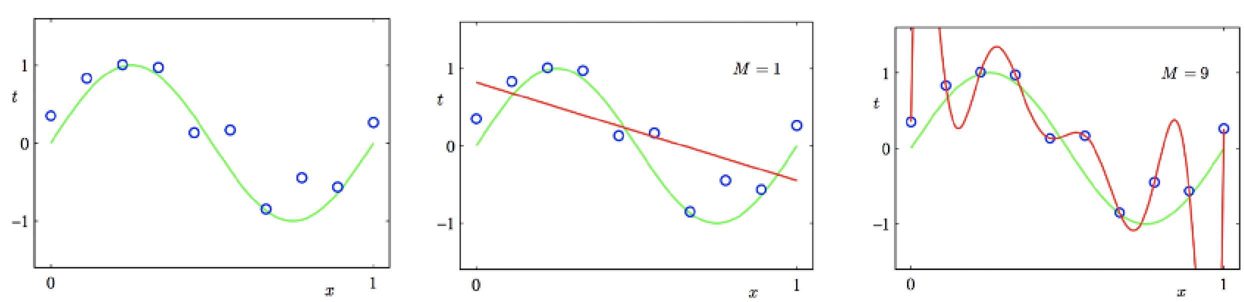
c) What's the difference between **validation** dataset and **test** dataset?

d) Why **accuracy** can be a bad metric to evaluate your classifier? What other metrics you can use?

e) Consider a synthetic data set which is generated from the function $\sin(2\pi x)$, which is shown by the green curve in Figure 2(a). Let us assume that we decided to fit a polynomial function of the following form:

$$y(x, \boldsymbol{w}) = w_0 + w_1 x + w_2 x^2 + ... + w_M x^M$$

1. Figure 2(b) shows what happens when we fit a linear model (M = 1 in the Equation) to the observations. Is this model a good fit? If not, is it underfitting or overfitting?
2. Figure 2(c) shows what happens when we fit a higher order polynomial (degree = M = 9 in the Equation) to the observations. Is this model a good fit? If not, is it underfitting or overfitting?



(a) Dataset (b) Model 1 (c) Model 2

Figure 2: Question 4(e)